

COM3502/4502/6502 SPEECH PROCESSING

Lecture 3

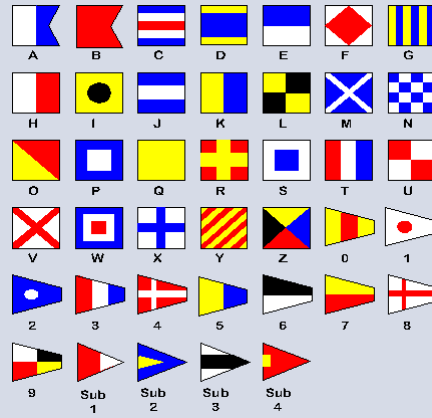
Signals

What is a Signal?



- A signal is a physical quantity that can carry '**information**'
- Discrete signals are selected from a finite alphabet or '**code book**'
- Continuous signals derive from ...
 - a *single*-dimensional '**scalar**' measurement
 - a *multi*-dimensional '**vector**' of measurements

Discrete Signals (Codes)

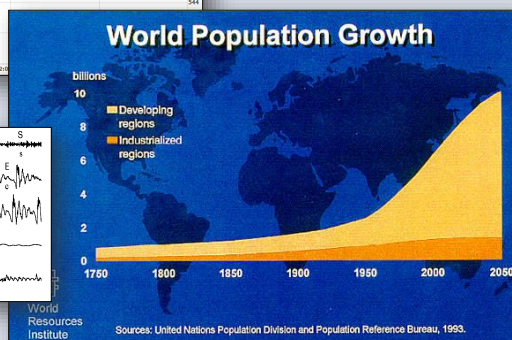
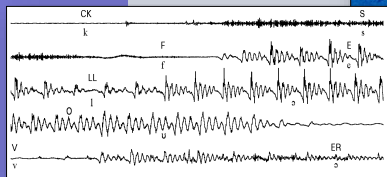


hello

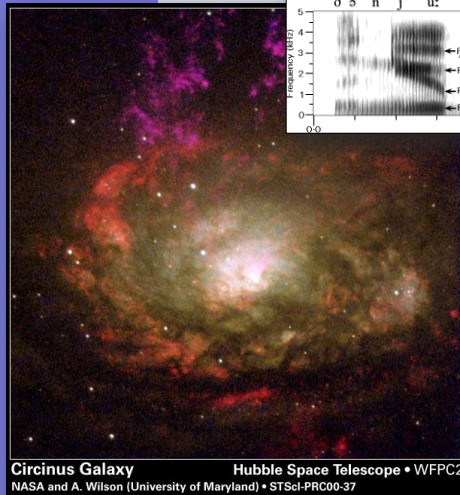
Continuous Scalar Signals



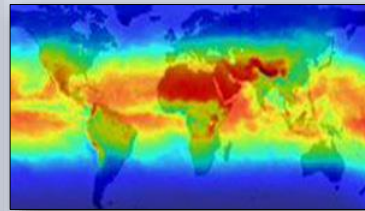
- Often depending on time: $x(t)$



Multidimensional Vector Signals



Circinus Galaxy Hubble Space Telescope • WFPC2
NASA and A. Wilson (University of Maryland) • STScI-PRC00-37

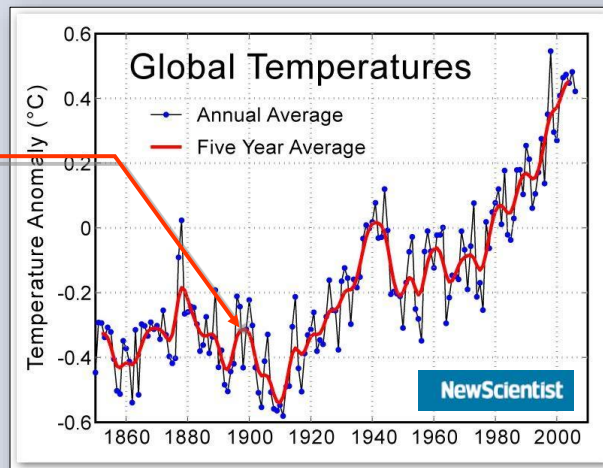


Signal Processing

- ‘**Signal processing**’ is the analysis, interpretation and manipulation of *sensor data*
- E.g., signal processing can be used to ...
 - compute global ‘**statistics**’
(means, modes, minima, maxima, etc.)
 - find ‘**patterns**’ in the data
(trends, repeats, etc.)
 - deduce ‘**causes**’ of the data
(e.g. conditioning variables)
 - predict ‘**outcomes**’ from the data
(e.g. the weather forecast)
 - ‘**transform**’ the data
(scaling, rotation, mapping, etc.)

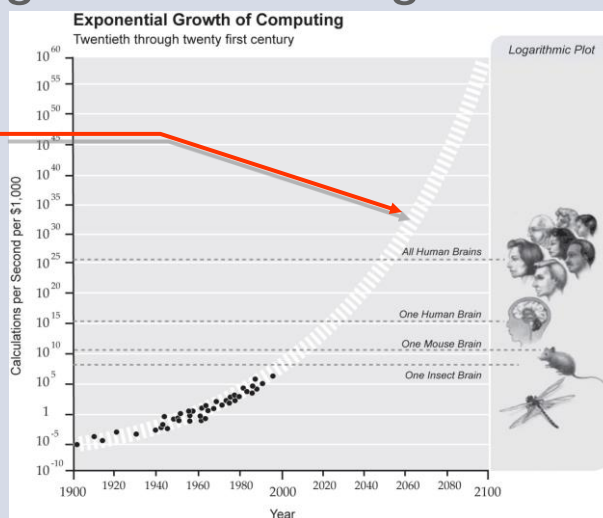
Signal Processing - Example

Trend line
(windowed
average)



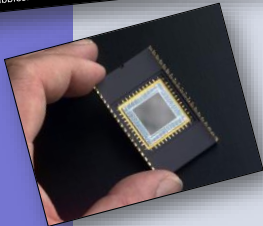
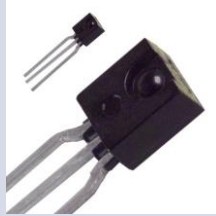
Signal Processing - Example

Prediction
(extrapolation)

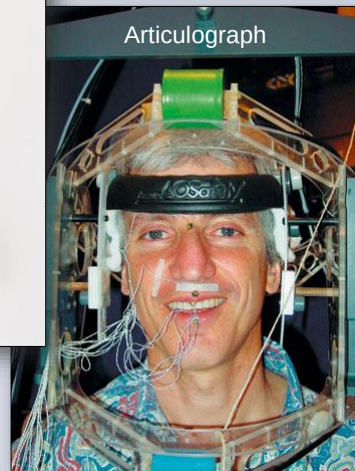
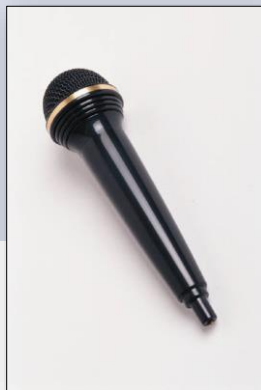


http://nyinquirer.typepad.com/nyinquirer/2006/08/human_20_kurzwe.html

Sensors/Transducers

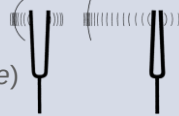


Sensors for Speech Signals



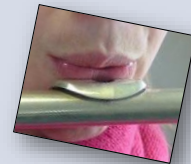
Acoustic Signals

- Physical systems can ...
 - respond to changes in air pressure (e.g. a microphone)
 - generate changes in air pressure (e.g. a loudspeaker)
- The mechanical properties of physical systems can cause them to ...
 - '**oscillate**' (e.g. a tuning fork)
 - '**resonate**' (e.g. an organ pipe)
- The physical dimensions determine the behaviour:
 - a long string vibrates at a lower frequency than a short string (e.g. on a stringed musical instrument)
 - a large cavity resonates at a lower frequency than a small cavity (e.g. in a wind musical instrument)



Acoustic Signals

- When *energy* is supplied continuously, then a resonant oscillation can be sustained, e.g. ...
 - blowing into a wind instrument
 - bowing a stringed instrument
 - whistling
 - voicing into a vocal tract
- When *energy* is supplied for a limited time period, then a resonant oscillation will be initiated and then decay over time, e.g. ...
 - plucking a stringed instrument
 - hitting a percussion instrument
 - a pulse from the vocal cords

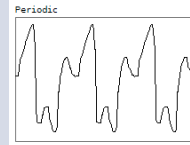
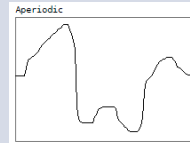


Types of Signal

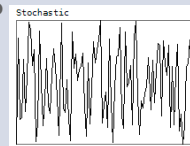
- Deterministic signals
(fully predictable, given the generator)

- ‘aperiodic’ signals

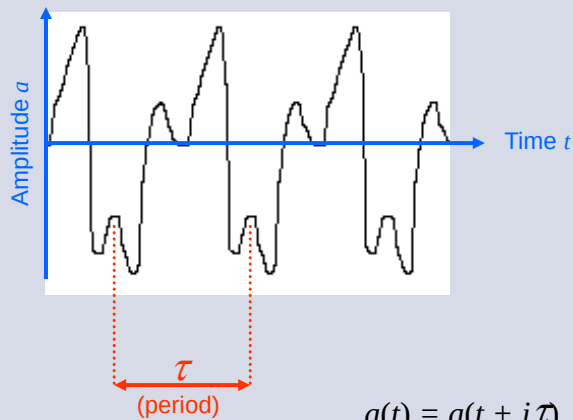
- ‘periodic’ signals



- Non-deterministic signals
(a function of a random variable,
so not fully predictable)



Periodic Signals

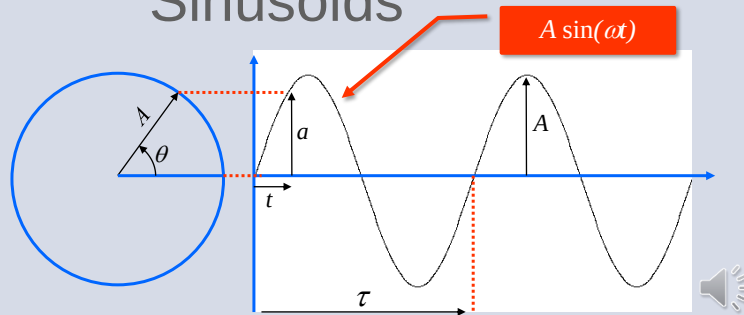


$$a(t) = a(t + i\tau)$$

for all t and integer i

Sinusoids

Projection of a fixed length vector rotating at a constant velocity



A radius = length of rotating vector = peak amplitude

θ angle (radians)

$f = 1/\tau$ frequency (cycles per second, Hertz)

$\omega = 2\pi f$ angular frequency (radians/second)

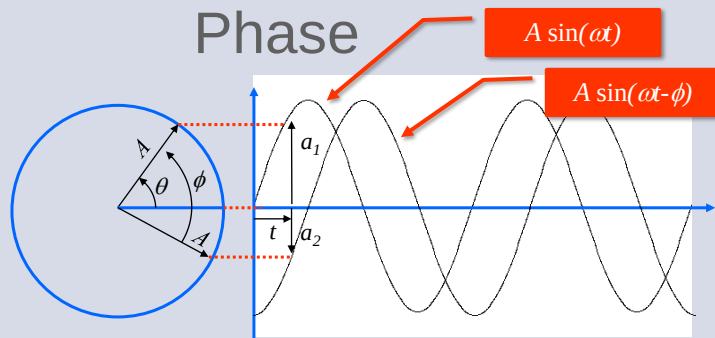
$\theta = \omega t$

$a = A \sin(\theta) = A \sin(\omega t)$

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Phase

Two sinusoids with equal frequency and amplitude may still be different in their 'phase'



A radius = length of rotating vectors = peak amplitudes

θ phase angle of 1st vector (radians)

ϕ phase lag/delay of 2nd vector wrt 1st (radians)

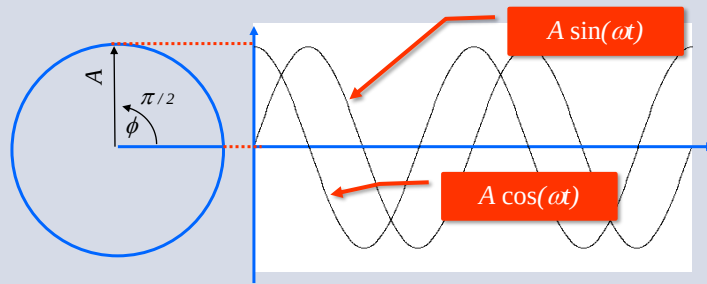
$a_1 = A \sin(\theta) = A \sin(\omega t)$

$a_2 = A \sin(\theta - \phi) = A \sin(\omega t - \phi)$

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Sine and Cosine Waves

The special case of a 90° (orthogonal) phase angle



A radius = length of rotating vector = peak amplitude

$\theta = 0$ phase angle of 1st vector (radians)

$\phi = \pi/2$ phase angle (radians)

$A \sin(\theta) = 0$

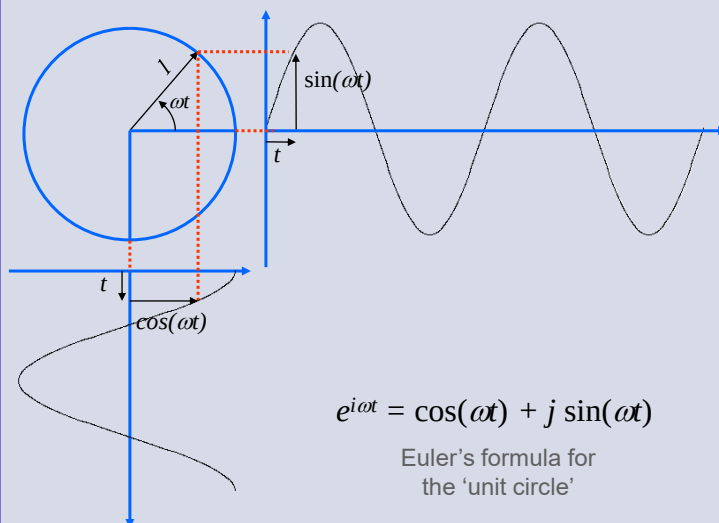
$A \sin(\theta + \phi) = 1$

$A \sin(\omega t + \pi/2) = A \cos(\omega t)$

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Sine and Cosine Waves

Orthogonal projections of a rotating unit vector



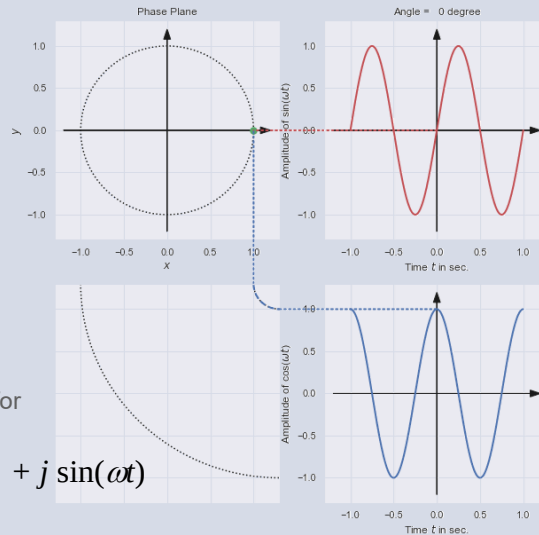
$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

Euler's formula for the 'unit circle'

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Sine and Cosine Waves

Orthogonal
projections of a
rotating unit
vector



Euler's formula for
the 'unit circle'

$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

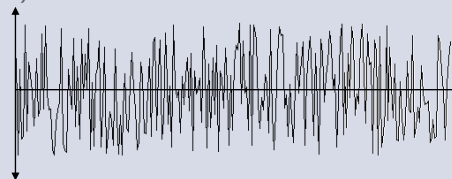
Source code: <https://github.com/Stefan-Goe/Speech-Processing-Notebooks/blob/main/signals/Signals.ipynb>

COM3502-4502-6502 Speech Processing: Lecture 3, slide 23

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White Noise

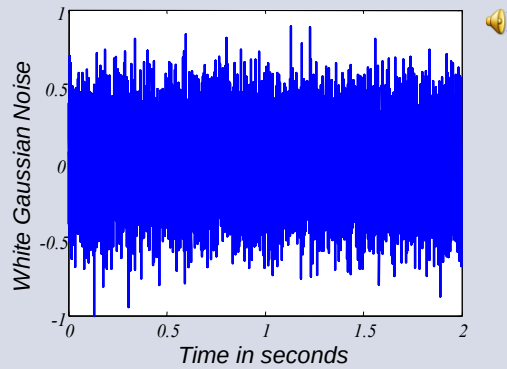
- Completely random '**stochastic**' signal
- The value of the signal at time t gives no information about the value of the signal at time $t + \delta t$ (i.e. they are completely '**uncorrelated**')
- White noise is characterised by ...
 - mean value μ
 - standard deviation σ (or variance σ^2)
- White noise can follow an arbitrary '**noise distribution**' ...
 - uniform distribution
 - Gaussian (*normal*) distribution



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White Noise

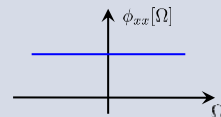
- Time signal of white Gaussian noise



White Noise

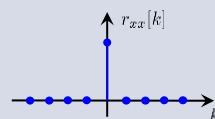
- Described by the constant Power Spectral Density

$$\phi_{xx}(\Omega) = \sum_{k=-\infty}^{+\infty} r_{xx}[k]e^{-j\Omega k} = \sigma_x^2$$



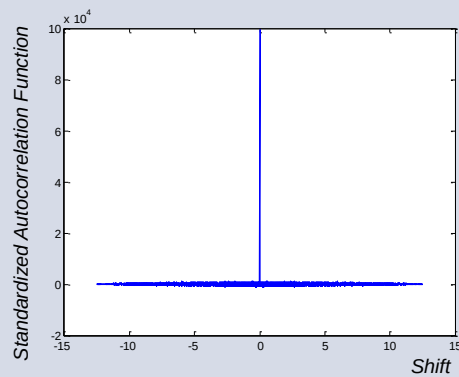
- Autocorrelation function:

$$r_{xx}[k] = E\{x^*[k]x[k+\kappa]\} = \sigma_x^2\delta[k] = \begin{cases} \sigma^2 & \text{for } k = 0 \\ 0 & \text{for } k \neq 0 \end{cases}$$



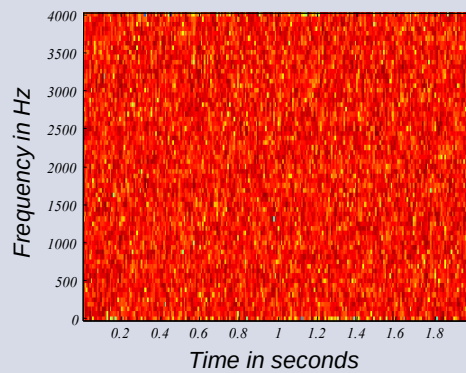
White Noise

- Autocorrelation function of white noise (estimated value)



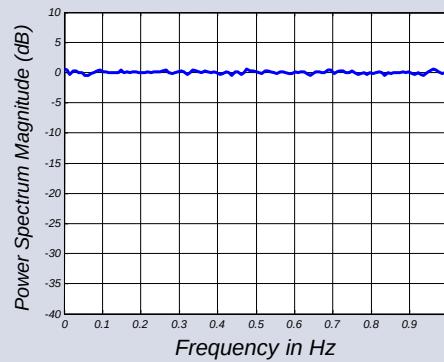
White Noise

- Spectrogram of White Noise



White Noise

- Power Spectral Density of white noise (estimated value)



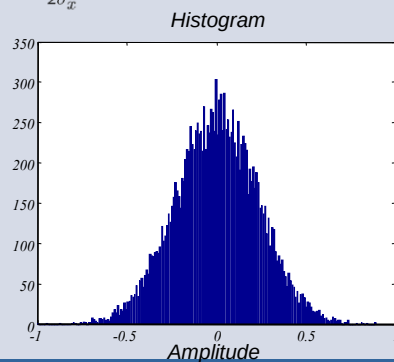
Gaussian White Noise

- Real Gaussian process
 - Identified by distribution of the **amplitude**
 - Probability density function:

$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma_x} e^{-\frac{(x-m_x^{(1)})^2}{2\sigma_x^2}}$$

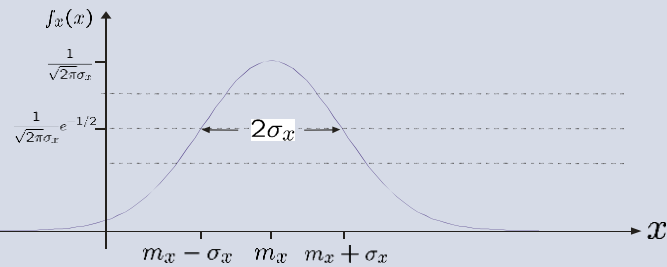
- Mean value:
 $m_x^{(1)}$

- Variance:
 σ_x^2



Gaussian Stochastic Process

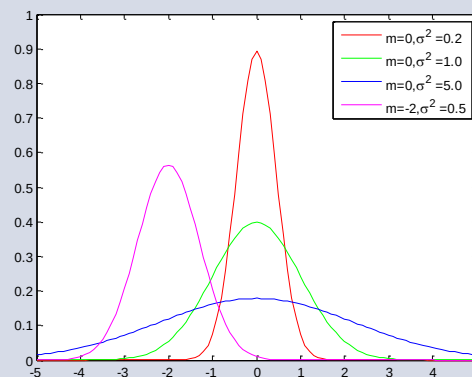
- Probability density:



Probability density is completely described by the Mean value m_x and the Standard deviation σ_x

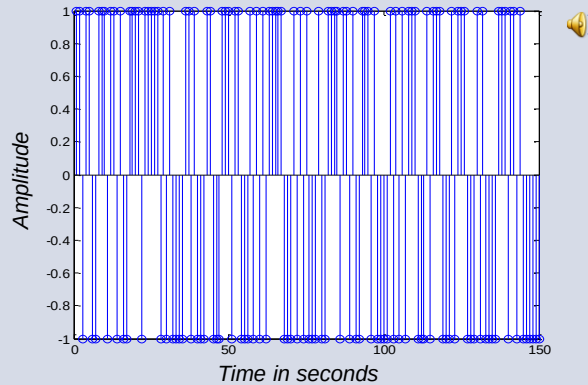
Gaussian Stochastic Process

- Probability densities of Gaussian processes:

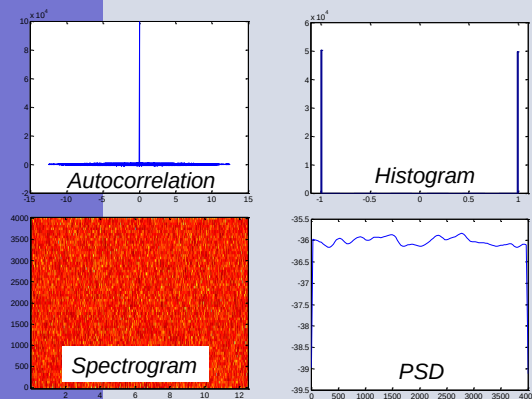


Binary Noise

- Binary White Noise



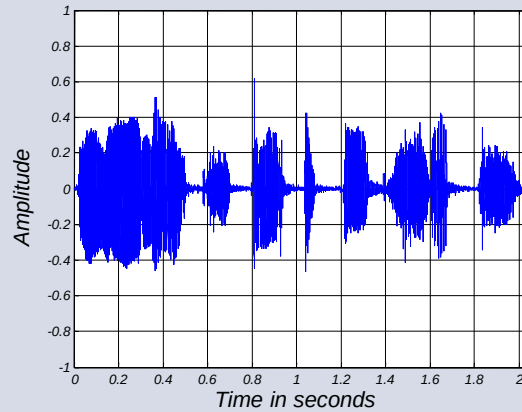
Binary (White) Noise



- Spectral and correlation properties hardly differ from white Gaussian noise.
- Different distribution.

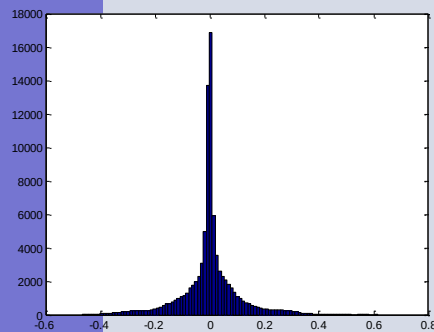
Speech

- Speech - time domain signal / wave form

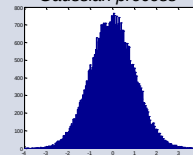


Speech

- Histogram of Speech



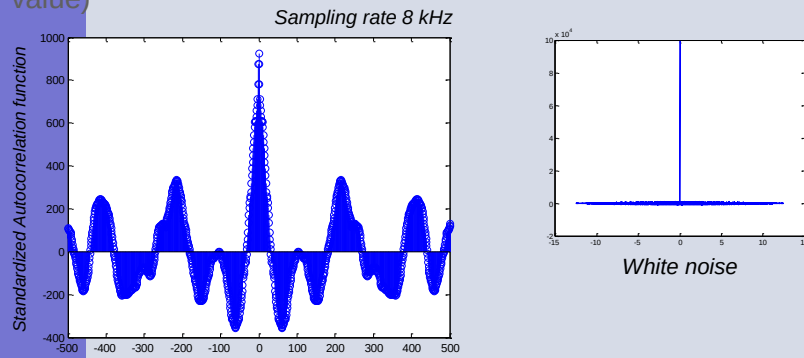
Gaussian process



- Super-Gaussian distribution
 - Laplace distribution

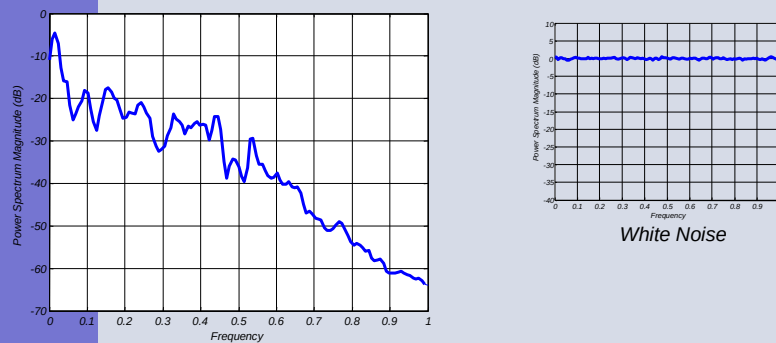
Speech

- Mean autocorrelation function of speech (estimated value)



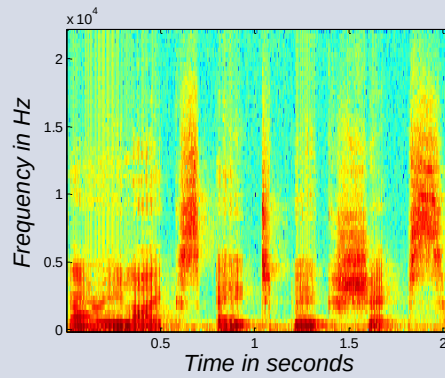
Speech

- Mean power spectral density of speech (estimated value)



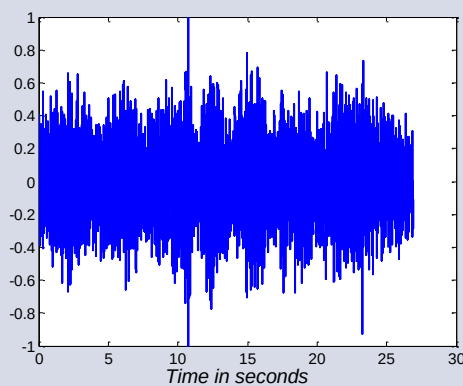
Speech

- Spectrogram of Speech (estimated value)



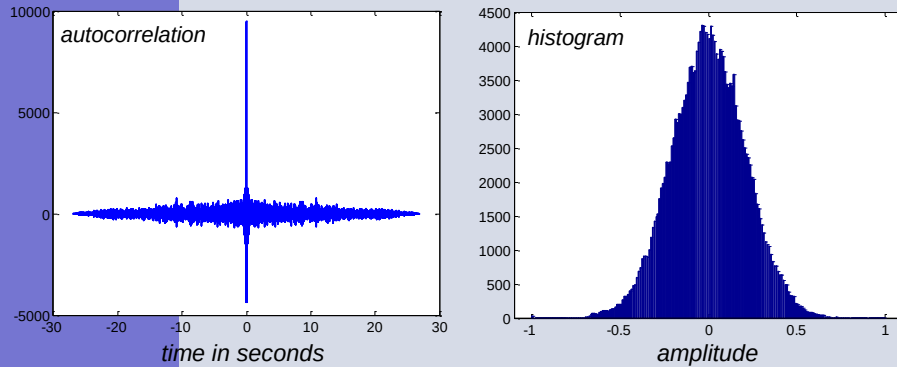
Car Noise

- Car Noise



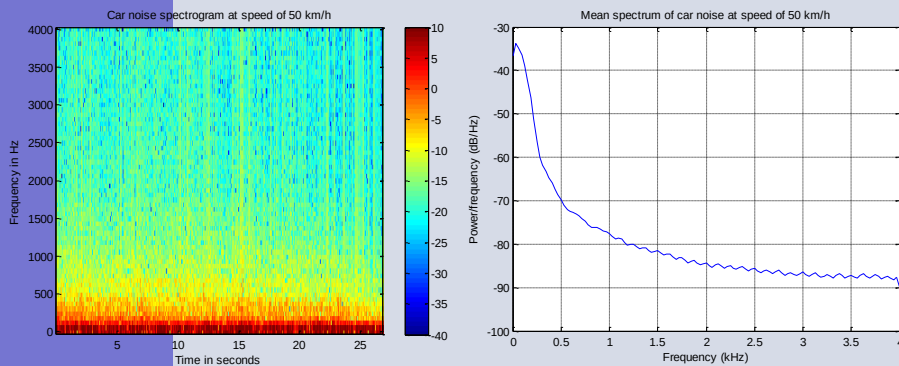
Car Noise

- Autocorrelation function of car noise (estimated value)



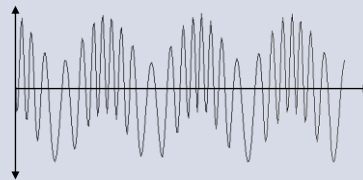
Car Noise

- Spectral properties of car noise
 - Wind, roll and engine noises, constant engine speed



Stationary Signals

- A '**stationary signal**' is one whose properties *do not* vary with time ...
 - a sine wave of constant amplitude and frequency
 - noise of constant mean and variance
- '**Non-stationary signals**' change with time ...
 - amplitude modulation
 - frequency modulation
 - modulated mean (*and variance*)



Non-Stationary Signals

- *Stationary* signals imply that there is a constant source of energy into the system
- Signals will be *non-stationary* if the energy in a system decays
- For example, a 'damped' sine wave ...

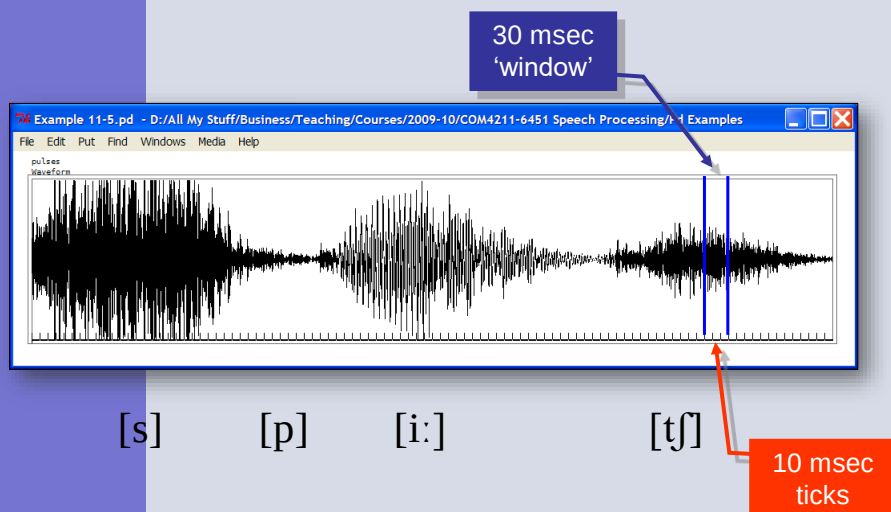
$$a = e^{-At} \cdot \sin(\omega t)$$



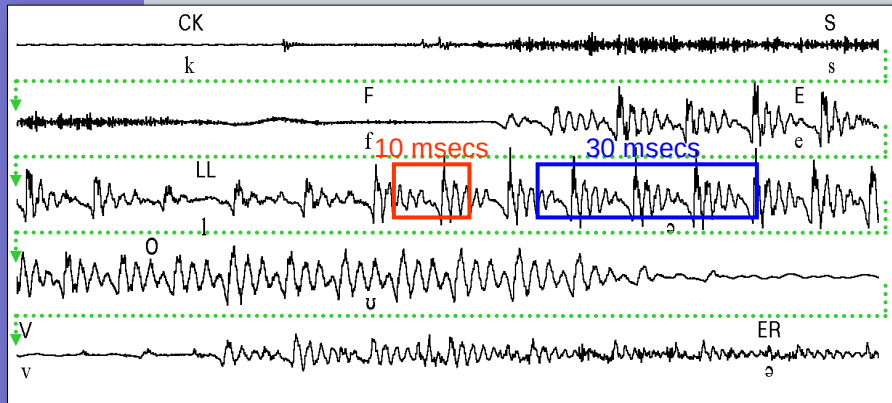
Quasi-Stationary Signals

- Speech is a mixture of periodic, aperiodic and stochastic signals, hence it is *non-stationary* by nature
- In practice, it is normally assumed that speech is stationary over a short time interval (*10-30 msec*)
 - if the interval is too short, there is insufficient time to determine the signal properties accurately
 - if the interval is too long, the speech properties vary significantly
- This '**quasi-stationary**' assumption is used in nearly all speech signal processing (*despite being a gross approximation*)

The Quasi-Stationarity of “Speech”



The Quasi-Stationarity of Speech



“bricks FELL OVER”

Taken from 'Speech Synthesis and Recognition', J. Holmes & W. Holmes, Taylor & Francis, London and New York, 2001.

This lecture has covered ...

- Discrete/continuous signal
- Signal Processing
- Sensors/transducers
- Microphones
- Periodic signals (*sines & cosines*)
- White noise and 'other' signals
- Stationarity

Any Questions ?

Post in the Blackboard Discussion group if something comes to your mind in between lectures



Next time ...

Digital Signals